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Structural equation modeling of caring behaviors and anxiety in nursing students during internship: the mediating role of ethical climate

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Abstract

Background Internship in nursing education strengthens competence, confidence, and professional skills. It is essential to identify the clinical factors affecting anxiety and the ethical climate among nursing students in this era. This study aimed to (1) identify caring behaviors and ethical climate as predictors of nursing students' anxiety using hierarchical regression analysis, and (2) examine the mediating roles of students' age and ethical climate in the relationship between caring behaviors and anxiety during internship.

Methods This is a cross-sectional study. 225 nursing students in Shiraz and Larestan Universities of Medical Sciences were selected to participate in this study using convenience sampling. Spielberger State-Trait Anxiety Inventory, Caring Behaviors Inventory, and Hospital Ethical Climate Survey were used for data collection. SPSS macro-PROCESS was used for data analysis.

Results Nursing students reported moderate levels of state anxiety and mild levels of trait anxiety. The nursing students' caring behaviors and the hospital's ethical climate were the predictors of anxiety during the internship. Moreover, age, caring behaviors, and perception of hospital ethical climate accounted for an additional 23% of the anxiety variance. Furthermore, hospital ethical climate was a mediator in the correlation between caring behaviors and anxiety in nursing students ($\beta = -0.06$, $CI = -0.11 - -0.02$).

Conclusion These findings suggest that enhancing caring behaviors and fostering a positive hospital ethical climate might help reduce nursing students' anxiety during internships. The mediating role of the ethical climate highlights its importance in the relationship between caring behaviors and anxiety, underscoring the need for supportive clinical environments.

Keywords Anxiety, Ethics, Clinical; students, Nursing; care, Internship and residency

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Background

The internship usually occurs in the fourth year of nursing, after students have completed the required curriculum [1]. During their internship, nursing students work under the guidance of an expert nurse in a clinical setting, utilizing their knowledge and skills to provide patient care. This program is essential for their progression from nursing students to practicing as professional nurses. However, due to the gap between theory and practice, heavy workload, and lack of clinical knowledge and skill, nursing interns face significant challenges during their internship. While the duration of the internship can influence the growth of nursing students' caring abilities, it may also induce anxiety in challenging environmental conditions [2]. Consequently, gaining insight into the current circumstances and factors associated with anxiety during the internship is crucial, as it can aid students in successfully transitioning from nursing students to professional nurses [3].

Most nursing students showed anxiety in clinical settings. Half of the students reported mild to moderate and approximately one-fifth indicated severe anxiety [4]. A study reported that more than half of nursing students experienced mild anxiety in clinical training [5]. In a study on Iranian nursing students, it was mentioned that they had anxiety during the pre-internship exam [6, 7]. During the internship, nursing students might have anxiety as a common psychological issue [8]. Caring for terminally ill patients and realizing that these patients may not recover might lead to anxiety in nursing students [9]. Therefore, it seems crucial to assess the nursing students' anxiety and the factors associated with it during the internship.

One of these factors may be caring behavior. The authors hypothesized that the dimensions of caring behavior, including respect for others, communication, skill and knowledge, and attention to others' experiences in clinical environments, might affect the anxiety of nursing students during internship. Caring is an essential part of nursing, and nurses are responsible for improving the nursing profession by providing necessary caring behaviors [10]. If a model of nurse-patient interaction based on desirable care behaviors is implicated, the work environment and nurse-patient satisfaction can be improved [11]. However, neglecting the development of caring relationships can harm mental health and cause anxiety [12].

Watson states that when there is a disharmony between the mind, body, and spirit, between the real self and the perceived self, a gap and disproportion occur, which leads to threat, anxiety, and inner turmoil [13]. Watson's theory (Theory of Human Caring), which is based on care, posits that maintaining human care is an important factor realized through interpersonal interactions; Watson suggests values relating to nursing care [14]. Previous studies

reported the effect of Watson's human care theory on nurses' caring behaviors and strengthening their empowerment, and the impact of this theory on improving mental health [15]. Based on this theory, one study reported that nurses' anxiety was correlated to caring behaviors [16].

Watson's Human Caring Theory rests on three core concepts: the interpersonal caring relationship, the care situation or moment of care, and the healing process, with the latter forming the central focus of Watson's Human Caring Theory. In practice, one of the nurse's essential responsibilities is to create a therapeutic environment that facilitates patients' natural capacity to heal, fosters caring relationships, and supports recovery [17].

From an organizational standpoint, the characteristics of such an environment, including hospitals, extend beyond physical attributes to encompass ethical and moral considerations. An atmosphere that consistently upholds respect, fairness, and compassion represents a positive ethical climate, which sustains the curative processes by aligning care delivery with ethical principles, promoting mutual trust, and safeguarding the moral well-being of both patients and caregivers [17, 18]. In this way, the improved environment, such as a hospital described in Watson's Human Caring Theory, can be understood as closely related to the concept of ethical climate, as both create conditions in which caring relationships and quality patient outcomes can be achieved.

Caring behavior is inherently grounded in ethical principles. To foster such behavior, a supportive hospital ethical climate is essential [19]. The hospital's ethical climate is usually evaluated through nurses' perceptions of how well they interact with their peers, patients, managers, and physicians, and how the hospital's vision and objectives are aligned with their professional values [20]. In this regard, a study has shown that, during the internship, nursing students can state their stress and anxiety without fear in a suitable ethical climate [21]. One study reported that workplace ethical climate was associated with anxiety [22]. Another study indicated that when a positive ethical climate prevailed in the organizational environment, job satisfaction improved, in turn providing a safe environment for quality healthcare [23, 24]. A university environment, such as events and initiatives, may affect nursing students' mental health [25]. On the other hand, a negative perception of the ethical climate predicts the risk of damage to employees' health [26]. In the context of nursing education, a stressful clinical learning environment causes anxiety and fear, which negatively affect learning outcomes [27, 28]. Although earlier research has examined the ethical climate as a mediator between ethical and caring behaviors [29, 30], there have been few investigations, especially on a global scale, into

its potential role in connecting caring behaviors to anxiety among nursing students during their internships.

To our knowledge, no study, either in Iran or other countries, has investigated the mediating role of ethical climate in the relationship between caring behavior and anxiety among nursing students during their internships, particularly within community-based care models. Therefore, based on previous studies and Watson's theory of Human Caring, the present study aimed to address this gap. To improve the evidence-based education and practice, the following hypotheses were posed in this study:

- 1) There is a significant association between caring behaviors, hospital ethical climate, and anxiety among nursing students during their internship.
- 2) Caring behaviors and hospital ethical climate significantly predict nursing students' anxiety, as shown through hierarchical linear regression analysis.
- 3) Age and hospital ethical climate mediate the relationship between caring behaviors and anxiety among nursing students during their internship.

Methods

Study design

This is a cross-sectional study.

Setting

This study was conducted at the School of Nursing at Shiraz University of Medical Sciences (SUMS) and Larestan University of Medical Sciences.

Participants

Nursing students in the School of Nursing at SUMS and Larestan University of Medical Sciences participated in the present study. The inclusion criteria of this study were being a nursing student in the seventh and eighth semesters, having been enrolled at one of the two selected schools since at least the sixth semester, and having completed at least one semester of the internship program. Nursing students who self-reported a diagnosed psychological disorder were excluded from the study.

Sample size

Based on a pilot study on twenty nursing students who had passed the internship program, $r^2 = 0.25$, $\alpha = 0.05$, and $1 - \beta = 0.90$, number of predictors in set A (caring behaviors items = 42), and number of predictors in set B (hospital ethical climate items = 26), using (<https://www.danielsoper.com/statcalc/calculator.aspx?id=16>), the minimum sample size to require the effect was estimated 223. A total of 250 questionnaires were distributed among nursing students although the target sample size was 223. Of these, 225 questionnaires were completed and returned,

while 25 were excluded due to withdrawal or non-response, resulting in an attrition rate of 10.0%. The final analysis was conducted on the 225 completed responses.

Data collection and sampling

The data were collected using convenience sampling. The research assistant met the students in the clinical environment and provided them with the questionnaires. Then, the students filled them out in the presence of the researcher's assistant.

Measures

Data were collected using a demographic questionnaire that assessed the nursing students' age and gender. Moreover, Spielberger State-Trait Anxiety Inventory (STAI), Caring Behaviors Inventory, and Hospital Ethical Climate Survey were used.

STAI consists of 40 items, with 20 items measuring state anxiety and 20 items measuring trait anxiety. Each item was scored using a 4-point Likert scale from "not at all" = 1 to "very much so" = 4. Therefore, the scores range from 40 to 160, and higher scores show more anxiety. The State and Trait Anxiety Inventory scores, which range from 20 to 80, are classified into three levels: mild [20–39], moderate (40–59), and severe (60–80) [31]. In this study, the internal consistency of the inventory was approved using Cronbach's $\alpha = 0.93$.

The Caring Behaviors Inventory consists of 42 items. It was designed by Wolf et al. and contains subscales including respecting others, ensuring human presence, having positive communication and orientation, having professional knowledge and skill, and paying attention to the experience of others. Each item of this inventory is scored using a 4-point Likert scale from "never" = 1 to "always" = 6. The total score of this inventory ranges from 42 to 252. Higher scores indicate higher caring behaviors [32]. For better comparability, the scores of each subscale and the total scores are standardized by dividing them by the number of items. The Persian version of the 16-item Caring Behaviors Inventory demonstrated good reliability and confirmed construct validity [33]. The internal consistency of the inventory in our study was approved using Cronbach's α (0.98). Cronbach's α for respecting others, ensuring human presence, having positive communication and orientation, having professional knowledge and skill, and paying attention to the experience of others subscales were 0.95, 0.96, 0.94, 0.93, and 0.92, respectively.

The Hospital Ethical Climate Survey, designed by Olson, consists of 26 items. It contains five subscales and assesses the relationship between the participants and their peers, patients, managers, hospitals, and physicians. Each item was scored using a 5-point Likert scale from "almost never true" = 1 to "almost always true" = 5. For

each subscale, according to the number of items, scoring is different; thus, for better comparability, the scores of each subscale and the total scores are standardized by dividing them by the number of items. The scores of the Hospital Ethical Climate Survey range from 1 to 5. Higher scores indicate a more positive view of the hospital's ethical climate. The validity of the Ethical Climate Survey was confirmed by Olson, using construct validity with Lisrel software. Its reliability was reported using Cronbach's alpha of 0.91 [34]. The Persian version of the Hospital Ethical Climate Survey showed acceptable internal consistency, with Cronbach's alpha values ranging from 0.69 to 0.85 and an overall coefficient of 0.94. Factor loadings for all ethical climate items fell between 0.50 and 0.80, indicating a strong structural validity for the instrument [35]. In our study, the internal consistency of the inventory was approved using Cronbach's alpha (0.96). Cronbach's alpha for peers, patients, managers, hospitals, and physicians' subscales were 0.88, 0.91, 0.92, 0.85, and 0.87, respectively.

Ethical considerations

This study was approved by the Research Ethics Committees of Schools of Nursing and Midwifery, Management, and Medical Information Science at Shiraz University of Medical Sciences with the code of IR.SUMS.NUMIMG.REC.1402.027. The participation/non-participation of the nursing students was voluntary in this study. The questionnaires were coded, and they were anonymous. The purpose of this study, the number of items, and the time needed in this study were explained in a consent form and read verbally to the students, as well. They signed the consent form.

Data analysis

Using SPSS version 24, we determined the mean and standard deviation, frequency, and percentage. As a significant association was observed between the nursing students' age and anxiety in the results section, age was added to the model. The association between caring behaviors, hospital ethical climate, and age with anxiety

in nursing students was reported using the Pearson correlation coefficient. In this study, a hierarchical linear model was used to assess the predictors of nursing students' anxiety.

Moreover, the hypothesized model of anxiety among nursing students who had passed the internship program guided our analysis (Fig. 1). Based on this model, hospital ethical climate and age were considered as mediators. The nursing students' anxiety model was tested using the mediation model of the SPSS macro-PROCESS v4.2 by Andrew F. Hayes (Model 6). This model was used when there were two mediators in the relationship between the two variables. $p < 0.05$ was considered significant.

Results

Two hundred and twenty-five nursing students who had passed the internship period participated in this study. Of them, 113 (50.2%) and 112 (49.8%) were male and female, respectively. The mean age of the nursing students was 23.44 (SD = 2.13).

As Table 1 shows, the mean score of nursing students' anxiety during the internship was 86.32 (SD = 17.50). The mean state anxiety score was 44.28 (SD = 8.86), while the mean trait anxiety score was 41.80 (SD = 10.19). State anxiety was categorized as moderate, while trait anxiety was considered mild.

The mean scores of nursing students' caring behaviors and hospital ethical climate were 218.68 (SD = 34.86) and 3.65 (SD = 0.76), respectively. Their subscales are presented in Table 1. The highest and lowest mean scores of the nursing students' caring behaviors were related to "professional knowledge and skill" and "positive communication and orientation", respectively. In addition, the highest and lowest mean scores of the nursing students' views regarding ethical climate were related to "managers" and "hospital" subscales, respectively.

There was no significant difference between the genders regarding anxiety ($t = 0.85$, $p = 0.35$) and hospital ethical climate ($t = 1.50$, $p = 0.13$). However, the mean score of females' ($M = 225.29$ (SD = 25.13)) caring behaviors was

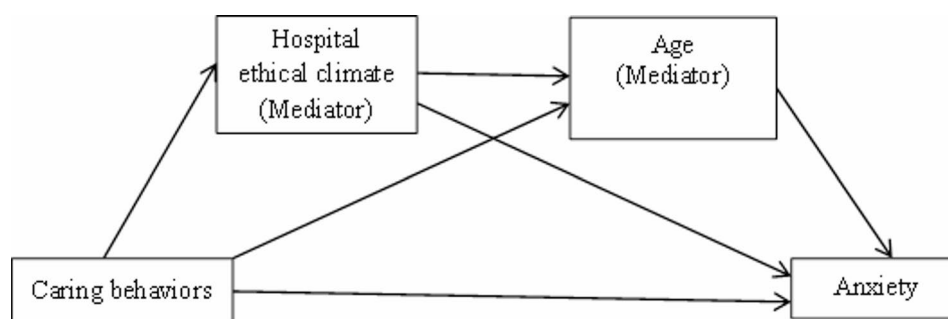


Fig. 1 The mediation roles of nursing students' age and hospital ethical climate in the relationship between caring behaviors and anxiety

Table 1 The mean score of the nursing students' anxiety and caring behavior and their views regarding hospital ethical climate during the internship

Variables	Minimum	Maximum	Mean (SD)	Standardized mean (SD)
Anxiety	47.00	144.00	86.32 (17.50)	-----
State anxiety	26.00	72.00	44.28 (8.86)	-----
Trait anxiety	21.00	77.00	41.80 (10.19)	-----
Caring behavior	42.00	252.00	218.68 (34.86)	5.20 (0.83)
Respect others	12.00	72.00	62.45 (10.14)	5.20 (0.84)
Ensuring human presence	12.00	72.00	62.57 (10.160)	5.21 (0.84)
Positive communication and orientation	9.00	54.00	45.80 (8.05)	5.08 (0.84)
Professional knowledge and skill	5.00	30.00	26.26 (4.17)	5.25 (0.83)
Pay attention to the experience of others	4.00	24.00	20.73 (3.55)	5.18 (0.88)
Hospital ethical climate	1.00	5.00	94.99 (19.76)	3.65 (0.76)
Patients	1.00	5.00	15.02 (3.54)	3.75 (0.86)
Peer	1.00	5.00	14.90 (4.45)	3.72 (0.86)
Hospital	1.00	5.00	20.25 (5.10)	3.37 (0.85)
Physicians	1.00	5.00	20.68 (5.17)	3.44 (0.86)
Managers	1.00	5.00	23.86 (5.28)	3.97 (0.88)

significantly higher than males ($M = 225.29$ ($SD = 25.13$), $t = -2.74$, $p = 0.007$).

A significant, negative, but weak association was observed between the nursing students' age and anxiety ($r = -0.13$, $p = 0.04$) (Table 2). Older nursing students experienced lower anxiety. However, there was no association between age and caring behaviors, and hospital ethical climate and their subscales ($p > 0.05$).

Caring behaviors and their subscales were significantly and negatively associated with anxiety ($p < 0.05$); however, the correlations of some subscales were weak ($r < 0.30$) (Table 2). This suggests that while increased caring behaviors are linked to lower anxiety levels, the strength of this relationship was modest and might have limited predictive power. Moreover, a significant, negative, and moderate association ($r = -0.44$ to -0.43) was observed between the hospital ethical climate (and its subscales) and anxiety ($p < 0.05$). This indicates that more positive perceptions of the hospital ethical climate are moderately associated with lower anxiety among nursing students, suggesting a potentially meaningful relationship in clinical or educational settings.

Normality assumptions for all variables (caring behavior, hospital ethical climate, and anxiety) ($p > 0.05$) and

regression residuals ($p = 0.08$) were confirmed using the Kolmogorov-Smirnov test, supporting the use of parametric analyses. The Durbin-Watson statistic of 1.75 indicated an absence of autocorrelation in the residuals, as it falls within the acceptable range of 1.5 to 2.5. Additionally, multicollinearity diagnostics demonstrated that all Variance Inflation Factor (VIF) values were below 10, confirming no significant multicollinearity among the independent variables (Table 3).

Hierarchical linear regression was conducted to identify predictors of nursing students' anxiety during the internship. In Step 1, age was not a significant predictor ($\beta = -0.12$, $p = 0.08$). In Step 2, both age ($\beta = -0.14$, $p = 0.03$) and caring behaviors ($\beta = -0.32$, $p < 0.001$) significantly predicted anxiety, with caring behaviors contributing an additional 10% to the explained variance. In Step 3, age ($\beta = -0.13$, $p = 0.04$), caring behaviors ($\beta = -0.18$, $p < 0.001$), and students' perceptions of the hospital ethical climate ($\beta = -0.36$, $p < 0.001$) significantly predicted anxiety, with ethical climate accounting for an additional 11% of the variance. Overall, the final model explained 23% of the variance in students' anxiety levels (Table 3).

The mediator effect of ethical climate.

Based on the mediators model (Table 4; Fig. 1), the overall model was significant, explaining 23% of the variance of nursing students' anxiety ($R = 0.47$, $R^2 = 0.23$, $MSE = 260.16$, $F = 18.85$, $p < 0.001$). Nursing students' caring behaviors had a significant direct effect on anxiety ($\beta = -0.09$, $p = 0.009$), as did perceptions of the hospital ethical climate ($\beta = -8.78$, $p < 0.001$) and age ($\beta = -1.21$, $p = 0.04$).

A mediation analysis revealed that the relationship between caring behaviors and anxiety was significantly mediated by students' perceptions of the hospital's ethical climate ($\beta = -0.06$, 95% CI $[-0.11, -0.02]$). However, no significant mediation effects were found for age, either alone or in combination with the hospital's ethical climate (Table 4).

Discussion

The nursing students experienced moderate levels of state and mild levels of trait anxiety during the internship period. A study on nursing students in China indicated that the anxiety level was moderate after experiencing the internship. The researchers pointed out that moderate anxiety had a protective role and could increase a person's ability to react to stimuli [6].

Our study showed that the nursing students' mean scores of state and trait anxiety were 44.28 ± 8.86 and 41.80 ± 10.19 , respectively. A study on 154 Turkish nursing students indicated their state and trait anxiety were 42.6 ± 9.4 and 45.3 ± 8.5 at the end of the 4th year of education, respectively [36]. Comparison of the present study and the research conducted in Turkey using t-test

Table 2 The association among anxiety, caring behaviors, hospital ethical climate, and their subscales, and age in nursing students

	1	2	3	4	5	6	7	8	9	10	11	12	13
1	r	1											
	p-value												
	95% CI ^a												
2	r	-0.31**	1										
	p-value	<0.001											
	95% CI ^a	-0.46, -0.14	1										
3	r	-0.30**	0.97**	1									
	p-value	<0.001	<0.001										
	95% CI ^a	-0.45, -0.13	0.96, 0.98	1									
4	r	-0.32**	0.97**	0.93**	1								
	p-value	<0.001	<0.001	<0.001									
	95% CI ^a	-0.47, -0.16	0.95, 0.98	0.88, 0.95	1								
5	r	-0.29**	0.96**	0.93**	0.92**	1							
	p-value	<0.001	<0.001	<0.001	<0.001								
	95% CI ^a	-0.44, -0.12	0.95, 0.98	0.89, 0.96	0.86, 0.95	1							
6	r	-0.31**	0.93**	0.89**	0.89**	0.86**	1						
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001							
	95% CI ^a	-0.46, -0.15	0.88, 0.95	0.80, 0.93	0.80, 0.93	0.78, 0.91	1						
7	r	-0.25**	0.92**	0.88**	0.86**	0.89**	0.89**	1					
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001					
	95% CI ^a	-0.40, -0.07	0.88, 0.95	0.81, 0.93	0.78, 0.91	0.83, 0.94	0.83, 0.94	0.38**	1				
8	r	-0.44**	0.38**	0.37**	0.35**	0.36**	0.38**	0.38**	0.38**	1			
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001			
	95% CI ^a	-0.56, -0.28	0.17, 0.55	0.15, 0.53	0.15, 0.53	0.16, 0.54	0.18, 0.54	0.18, 0.54	0.18, 0.54	0.18, 0.54	1		
9	r	-0.41**	0.42**	0.41**	0.39**	0.41**	0.39**	0.39**	0.39**	0.39**	0.88**	1	
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	
	95% CI ^a	-0.54, -0.25	0.23, 0.58	0.22, 0.58	0.21, 0.56	0.23, 0.56	0.22, 0.55	0.22, 0.55	0.22, 0.55	0.22, 0.55	0.84, 0.92	0.85**	1
10	r	-0.41**	0.40**	0.39**	0.37**	0.38**	0.40**	0.40**	0.40**	0.40**	0.90**	0.85**	1
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
	95% CI ^a	-0.54, -0.25	0.22, 0.57	0.19, 0.57	0.18, 0.54	0.21, 0.55	0.19, 0.55	0.19, 0.55	0.19, 0.55	0.19, 0.55	0.86, 0.93	0.78, 0.90	1
11	r	-0.32**	0.28**	0.28**	0.26**	0.26**	0.30**	0.30**	0.30**	0.30**	0.91**	0.73**	0.81**
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
	95% CI ^a	-0.47, -0.16	0.08, 0.46	0.08, 0.46	0.06, 0.43	0.09, 0.45	0.11, 0.46	0.11, 0.46	0.11, 0.46	0.11, 0.46	0.88, 0.93	0.64, 0.80	0.74, 0.86
12	r	-0.34**	0.14*	0.16*	0.12	0.13	0.13	0.13	0.13	0.13	0.83**	0.63**	0.64**
	p-value	<0.001	0.04	0.02	0.09	0.06	0.04	0.05	0.05	0.05	<0.001	<0.001	<0.001
	95% CI ^a	-0.50, -0.18	-0.09, 0.35	-0.07, 0.36	-0.11, 0.32	-0.10, 0.33	-0.08, 0.33	-0.09, 0.33	-0.09, 0.33	-0.09, 0.33	0.77, 0.88	0.53, 0.72	0.54, 0.74
13	r	-0.44**	0.45**	0.43**	0.44**	0.43**	0.42**	0.45**	0.45**	0.45**	0.85**	0.75**	0.75**
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
	95% CI ^a	-0.56, -0.29	0.26, 0.61	0.22, 0.60	0.25, 0.60	0.24, 0.58	0.24, 0.58	0.28, 0.60	0.28, 0.60	0.28, 0.60	0.79, 0.89	0.66, 0.83	0.66, 0.83
											0.55, 0.74	0.43, 0.66	0.43, 0.66

Table 2 (continued)

	1	2	3	4	5	6	7	8	9	10	11	12	13
14	r	-0.13	-0.06	-0.04	-0.05	-0.08	-0.04	0.01	0.03	0.02	-0.03	0.02	0.03
	p-value	0.04	0.40	0.50	0.47	0.26	0.55	0.81	0.67	0.77	0.64	0.71	0.66
	95% CI ^a	-0.22, -0.003	-0.14, 0.01	-0.14, 0.03	-0.15, 0.03	-0.17, -0.009	-0.12, 0.03	-0.09, 0.10	-0.11, 0.14	-0.10, 0.12	-0.14, 0.06	-0.07, 0.12	-0.06, 0.11

1. Anxiety, 2. Caring behaviors, Dimensions of caring behaviors: 3. Respect others, 4. Ensuring human presence, 5. Positive communication orientation, 6. Professional knowledge and skill, 7. Pay attention to the experience others, 8. Hospital ethical climate, Dimensions of hospital ethical climate: 9. Patients, 10. Peer, 11. Hospital, 12. Physicians, 13. Managers, and 14. Age

^aCI/Confidence Interval

*Correlation is significant at the 0.05 level (2-tailed)

**Correlation is significant at the 0.01 level (2-tailed)

analysis showed that the state anxiety of the two studies was approximately similar. However, students in our study reported significantly lower levels of trait anxiety than those of the Turkish study. It seems that in our structured internship program, including close coordination between academic supervisors, clinical staff, and students, might have fostered a more supportive environment that helped the students manage long-term anxiety. Cultural and educational differences between the two settings may also be relevant, as variations in clinical education practices, student-teacher relationships, and institutional support systems can all influence how students experience and manage anxiety over time.

Consistent with our findings, previous research has identified that age is a predictor of anxiety [4]. In our study, older nursing students reported lower levels of anxiety. In contrast, research conducted among Turkish nursing students found that those in their fourth year of study had higher mean scores for both state and trait anxiety compared with first-year students [36].

This study showed the mean score of the nursing students' caring behaviors was 218.68, and the standardized score was 5.20 ± 0.83 . A study on Turkish nursing students reported that the standardized score of nurses' caring behaviors was 5 ± 1.1 [37]. Another study on Iranian nursing students reported that internship was effective in improving academic, professional, and clinical skills. It was shown that during the internship period, the nursing students' searching in sources and patients' records, self-learning, reading textbooks, and searching to answer the academic supervisors' questions led to improved knowledge and skills [38]. Moreover, our study indicated that the highest and lowest mean scores of nursing students' caring behaviors were related to "professional knowledge and skill" and "positive communication and orientation", respectively. These findings can be interpreted through Watson's Theory of Human Caring, which emphasizes both technical competence and meaningful human interaction [39]. The highest scores in "professional knowledge and skill" suggest strong development in clinical competence, while lower scores in "positive communication and orientation" may reflect the need for further focus on the relational aspects of care. This indicates that while the internship period supports skill development, additional strategies may be needed to enhance interpersonal and emotional dimensions of caring.

The findings of this study showed that the standardized mean score of the hospital ethical climate, from the perspective of internship students, was 3.65. In addition, the highest mean score belonged to the relationship between the nursing students and their managers. It means that the nursing students had a positive view of their managers. In a study conducted in Flanders, the researchers reported that the standardized mean of this variable from

Table 3 Hierarchical linear regression analysis of age, caring behaviors, and hospital ethical climate in prediction of nursing students' anxiety during internship

	Step 1					Step 2					Step 3				
	β	t	p	95% CI	VIF ^a	β	t	p	95% CI	VIF ^a	β	t	p	95% CI	VIF ^a
Constant		7.17	<0.001	82.17–144.50			8.91	<0.001	120.10–188.38			10.23	<0.001	135.23–199.84	
Age	–0.12	–1.73	0.08	–2.48–0.16	1.00	–0.14	–2.11	0.03	–2.60––0.09	1.004	–0.13	–2.03	0.04	–2.38––0.03	1.006
Caring behaviors						–0.32	–4.70	<0.001	–0.23––0.09	1.004	–0.18	–2.63	0.009	–0.16––0.24	1.177
Hospital ethical climate											–0.36	–5.23	<0.001	–12.09––5.47	1.173
R	0.12					0.35					0.48				
R ²	0.01					0.12					0.23				
Adjust R ²	0.01					0.11					0.22				
R ² change	0.01					0.10					0.11				
F change	3.00					22.12					27.43				
p-value F change	0.08					<0.001					<0.001				
VIF Variance Inflation Factor															
^a Collinearity Statistics															

the nurses' perspective was 3.84 [40], which is similar to the score obtained in the present study. Notably, the nurses in Flanders' study had an average of 8 months of work experience, yet they had already developed an understanding of the hospital ethical climate. In our study, it was indicated that the nursing students were able to understand the hospital's ethical climate well during the internship period, approximately similar to 8 months of work experience as a nurse.

The results of the present study showed that during the internship period, nursing students who had lower mean scores of caring behaviors reported higher anxiety mean scores. Despite the weak correlations, caring behaviors explained an additional 10% of the variance in nursing students' anxiety. These findings may still hold clinical relevance, as even small improvements in caring behaviors could help reduce anxiety in high-stress educational and clinical environments, such as internships. Similarly, Yerkes-Dodson law stated, "Individuals with more anxiety tend to fall on the high-arousal/low-performance portion of the curve" [41]. In this regard, a qualitative study indicated that behavioral, cognitive, and physiological changes in anxiety led to worry and fear in nursing students during the internship [6]. Other researchers showed that teachers' caring behavior had a direct effect on the students' learning anxiety [42]. Another study on nurses working in COVID-19 wards indicated that caring behaviors were associated with their death anxiety [12]. Stress created by training activities in clinical setting might reduce the nursing students' quality of life [43]. It shows that student' caring behaviors, such as respecting others, paying attention to others' experiences, having communicating, and having professional knowledge and skill, affect anxiety as a psychological and behavioral state. Based on Watson's Theory of Human Caring, core aspects of caring—such as empathy, compassion, and concern for oneself and others—are influenced by individual backgrounds, including cultural values and early life experiences [39, 44]. These humanistic qualities can act as internal coping mechanisms, helping nursing students' better handle stress and anxiety in demanding clinical environments, like internships. Promoting caring behaviors through reflective learning and a supportive educational and ethical atmosphere may not only contribute to students' professional growth but also help reduce the anxiety they encounter in clinical practice.

The observed moderate, negative correlations between nursing students' anxiety and their perceptions of the hospital ethical climate suggest that a more positive ethical climate might play a role in reducing anxiety. While the strength of these associations was moderate (ranging from –0.31 to –0.43), they indicated a consistent pattern that could have practical significance in clinical education. Additionally, the hospital's ethical climate accounted

Table 4 The mediating effect of hospital ethical climate in the relationship between caring behaviors and anxiety the nursing students experinced in the internship program

	Coefficient	SE ^b	t Statistics	p-values	95% CI
Direct effect					
Caring behaviors -> Anxiety	-0.09	0.03	-2.63	0.009	-0.16- -0.02
Hospital ethical climate -> Anxiety	-8.78	1.67	-5.23	<0.001	-12.09--5.47
Age -> Anxiety	-1.21	0.59	-2.03	0.04	-2.38- -0.03
Caring behaviors -> Hospital ethical climate	0.008	0.001	5.60	<0.001	0.005-0.01
Caring behaviors -> Age	-0.004	0.004	-0.99	0.31	-0.13- 0.004
Hospital ethical climate -> Age	0.12	0.20	0.59	0.55	-0.28- 0.53
Indirect effects					
Caring behaviors -> Hospital ethical climate -> Anxiety	-0.06	0.02 ^c	----	----	-0.11- -0.02
Caring behaviors -> Age -> Anxiety	0.005	0.004 ^c	----	----	-0.0009-0.01
Caring behaviors -> Hospital ethical climate-> Age -> Anxiety	-0.001	0.001 ^c	----	----	-0.004-0.001

Hospital ethical climate and age as mediators

^bStandard Error^cBootstrapping Standard Error

for 11% of the variance in anxiety, which, although modest, points to a meaningful contribution. This finding underscores the potential value of fostering ethically supportive clinical environments to reduce students' anxiety during the internship period. A study reported a strong association between climate anxiety and environmental attitudes in nursing university colleagues [27]. Another study showed that climate change in anxiety predicted pro-environmental behaviors in adolescents [45]. Previous studies have underscored a notable relationship between workplace ethical climate and anxiety [22].

The results of this study showed that there was a relationship between the hospital's ethical climate and the caring behavior of nursing students during the internship. Similarly, a study reported that organizational climate was correlated to the perceived organizational performance of the employees [46]. Researchers indicated a relationship between ethical climate and the quality of care from the nurses' point of view [47]. Other researchers also stated that there was a relationship between moral sensitivity and the caring behavior of nurses [48]. According to researchers, when nurses have a high understanding of the hospital's ethical climate, they can adapt and judge their behaviors according to the hospital values [47]. The Watson's Human Caring Theory emphasizes the ethical and human-centered foundations of caring and healing, not only for the individual nurse but also for the organizational system, such as hospitals and those receiving care. By grounding practice in these moral principles, it fosters an ethical climate in which respect, compassion, and integrity guide interactions, thereby supporting both professional well-being and quality patient cares [49].

Our study indicated that the relationship between caring behaviors and anxiety was significantly mediated by nursing students' viewpoints of hospital ethical climate.

This showed that when a nursing student was performing caring behavior, if she/he had a better understanding of the hospital's ethical climate, her/his anxiety would be decreased. A systematic review showed that a positive perception of ethical climate affected job satisfaction [50]. Other researchers showed an association between ethical climate and caring efficacy mediated by ethical sensitivity [51].

Strengths and limitations

Using the hierarchical linear model in steps to determine anxiety predictor variables was one of the strengths of this study. It is a powerful tool for evaluating the correlation between a dependent variable (anxiety) and multiple independent variables (students' caring behaviors, hospital ethical climate, and age). In addition, using the mediation model to determine the role of the hospital's ethical climate and age as mediators was another strength in this study. This is a cross-sectional study, so we could not conclude causality, which was one of the limitations of this study. Convenience sampling from a single country may have introduced sampling bias, limiting representativeness and generalizability beyond the Iranian context. Self-reported data are vulnerable to recall errors, social desirability, and response bias. Potential confounding variables, such as prior clinical experience and exposure to challenging clinical situations, were not assessed.

Clinical and academic implementations

Given that students' anxiety correlated with caring behaviors and the hospital ethical climate during the internship, we recommend that academic and hospital administrators should focus on promoting caring behaviors, such as respecting others, acknowledging others' experiences, communicating effectively, and demonstrating professional knowledge and skill, and fostering a

positive ethical climate, as these may reduce the students' anxiety.

Considering that nursing students had high caring behaviors and a positive perception of the hospital's ethical climate during the internship period, it is suggested that the implementation of this program should continue in the nursing curriculum and other nursing educational centers to improve the students' caring behaviors. By using this program, we hope the nursing shortage will decline. We hope that the nursing shortage will be decreased through the use of this program.

Conclusion

Nursing students in this study reported moderate levels of state anxiety and mild levels of trait anxiety. The findings indicated that both students' caring behaviors and the hospital's ethical climate were significant predictors of anxiety during the internship period. Furthermore, the hospital's ethical climate was found to mediate the relationship between caring behaviors and anxiety. Based on these results, implementing this program for fourth-year nursing students is recommended, as it may enhance students' caring behaviors and help address, at least in part, the shortage of nursing staff.

Abbreviations

SUMS	Shiraz University of Medical Sciences
STAI	Spielberger State-Trait Anxiety Inventory
VIF	Variance Inflation Factor

Acknowledgements

The authors would like to thank the nursing staff and managers who collaborated in the implementation of the intervention and nursing students' education in clinical settings and Shiraz University of Medical Sciences. The authors thank Razieh Rasekh, Hamideh Falah, Laila Hashemizadeh, and Miss Azadi for collaboration in the implementation of the internship in Nemazee hospital. The authors would like to thank Shiraz University of Medical Sciences, Shiraz, Iran, and Center for Development of Clinical Research of Nemazee Hospital and Dr Nasrin Shokrpour for English language editorial assistance.

Authors' contributions

MR, NP, RD, AP, MSH participated in conceptualization of this study. RD participated in data collection. MR and AP conducted the management of the data analysis. MSH, RD, and MR participated in the intervention management. All authors participated in writing and approving the original draft of the manuscript.

Funding

Shiraz University of Medical Sciences (Grant Number = 28249) financially supported this study.

Data availability

By email to Masoume Rambod, the data of this study would be available.

Declarations

Ethics approval and consent to participate

This study was approved by Research Ethics Committees of Schools of Nursing and Midwifery, Management and Medical Information Science-Shiraz University of Medical Sciences (IR.SUMS.NUMIMG.REC.1402.027). Participation in the study was optional for nursing students. The study was conducted following the Declaration of Helsinki. There was no need to write names on the questionnaires, and they were coded. The consent to participate in this

study was obtained from the participants. All the nursing students signed the consent form. It was confirmed that this consent was informed.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 23 June 2024 / Accepted: 9 December 2025

Published online: 15 January 2026

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